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Identification of genomic regions for rust resistance in sorghum

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Summary

The location and effects of genomic regions for rust resistance in sorghum were determined. One hundred and sixty recombinant inbreds, which derived from a cross between QL39 and QL41, were used as a segregating population for genome mapping and rust resistance evaluation. Phenotypic data were collected in replicated field trials in two years. Interval mapping and non-parametric mapping identified four regions, each in a separate linkage group, associated with rust resistance. The region with the largest effect on rust resistance is on linkage group 10; it accounted for 40% of the total phenotypic variation.

Introduction

Foliar diseases of sorghum are among the most common and recognisable diseases of sorghum [*Sorghum bicolor* (L.) Moench]. Sorghum rust (*Puccinia purpurea* Cooke) is widely distributed and occurs in almost all sorghum growing areas of the world. The disease is important because its presence predisposes sorghum to other major disease problems, such as the *Fusarium* stalk rots, charcoal rot, and grain molding (Frederiksen, 1986).

The genetics of host-pathogen interactions for sorghum rust have been investigated in several studies and the number of genes involved varied in different populations (Hooker, 1985). In crosses between the sweet sorghums Planter and MN 960, resistance to sorghum rust was controlled by the simple dominant gene *Pu* (Coleman & Dean, 1961). Different allelic interactions at the *Pu* locus have been reported (Miller & Cruzado, 1969). Segregation ranging from immune to highly susceptible, with intermediate classes of highly resistant, semi-resistant, and susceptible, were observed in progeny from a cross between a highly susceptible cultivar and a highly resistant cultivar in India (Patil-Kulkarni et al., 1972). It was concluded that more than one pair of genes were segregating in this population. In another study, resistance was believed to be due to three major genes (Rana et al., 1976). In all these studies, classical genetic analyses were applied to either test-crosses or F2/F3 segregating populations, and phenotypic scores for rust were classified in three groups based on the presence or absence of uredia.

Molecular markers, such as RFLPs, SSRs and RAPD, have been successfully employed to construct linkage maps and to locate genes with qualitative and quantitative effects. In sorghum, several linkage maps based on RFLP markers have been established (Paterson, 1994; Xu et al., 1994; Pereira et al., 1994; Chittenden et al., 1994; Dufour et al., 1997). Identification of genetic regions affecting morphological traits such as plant height (Pereira & Lee, 1995; Lin et al., 1995), maturity (Lin et al., 1995), and also agronomically important traits such as drought tolerance (Tuinstra et al, 1996) and insect resistance (Tao et al., 1997) have been reported.

In the investigation presented here, we report the detection of genomic regions with major effects on rust infection in sorghum using a recombinant inbred population and a genetic linkage map constructed on the population.

Material and methods

Plant material

One hundred and sixty random recombinant inbred lines (RILs) were developed from the cross QL39 × QL41. Both the parental lines and the RILs were developed by the Queensland Department of Primary Industries (Henzell, 1992; Henzell et al., 1994).

Field trials and phenotypic data collection

All RILs were planted at the Queensland Department of Primary Industries' Gatton Research Station in two seasons, 1994–1995 and 1995–1996, as single row plots (5.5 m × 0.7m) with two replications. Two field trials were conducted in the 1995–1996 season with two weeks difference in planting date. Phenotypic data for rust were collected at maturity. Rust infection was visually scored based on a 1–9 rating system, where a rating of 1 indicates no infection on plant leaves, and 9 indicates very heavy infection.

Linkage map

The genetic linkage map developed on the same RIL population was used in this study. A total of 166 loci were mapped onto 21 linkage groups with a total genetic distance of approximately 1400 cm. Linkage groups, locus order, and genetic distance were as described in a previous report (Tao et al., 1998).

Data analysis

MapQTL (Van Ooijen & Maliepaard, 1996) was used in this study. Both non-parametric mapping and interval mapping were conducted separately on each phenotypic data set to identify genomic regions associated with rust resistance. Non-parametric means that no assumptions are being made for the probability distribution(s) according to which the quantitative trait (after fitting the QTL phenotype) should behave. For the non-parametric mapping method, the rank sum test of Kruskal-Wallis is used in MapQTL. The test ranks all individuals according to the quantitative trait, and calculates a test statistic based on the ranks using the classification according to the genotypes. A stringent significant level of 0.005 was applied in this study, as suggested by the authors of the program to identify the loci with genetic effects on rust resistance. For interval mapping, an LOD threshold of 2.4 was used to declare the presence of a putative QTL in a given genomic region. This threshold level corresponds to a test of presence of a QTL at the 0.05 level of significance according to the 'sparse-map' case (Lander & Botstein, 1989; Van Ooijen, 1992).

Results

Phenotypic data

Phenotypic data for rust were collected from one trial in the 1994–1995 season, coded as R-95, and two trials in the 1995–1996 season, coded as R-961 and R-962 respectively. Rust infection levels ranged from plants having only a few uredia (score 1) to those having most of the leaf area covered with uredia (score 9). Both parental lines exhibited some resistance to rust in all three field trials, QL41 with high and QL 39 with intermediate levels of resistance. The frequency distribution of rust resistance scores for the RIL population showed a continuous distribution. Distributions of data R-95 and R-961 were skewed towards rust resistance, while the rust scores for R-962 was normally distributed. This implies that there was less infection in the former trials than the latter trial. Histograms of the frequency distribution for the rust scores in the three different experiments are presented in Figure 1.

QTL identification

Both non-parametric mapping and interval mapping were conducted on each data set separately. From interval mapping of the data from 1995, three regions, on linkage groups 4, 5, and 10, were identified with LOD scores higher than the threshold of 2.4. From data sets R-961 and R-962, interval mapping identified three unlinked genomic regions on linkage group 2, 5, and 10 with LOD scores higher than the threshold 2.4 (Figure 2).

In agreement with the results from interval mapping, the Kruskal-Wallis test (the no-parametric equivalent of one-way analysis of variance) identified loci in the same regions of linkage groups 2, 4, 5 and 10 from all three data sets. Loci BNL5.09 and TXS1625 in linkage group 2, and SSCIR51 and TXS1042 in linkage group 5 had significant effects on rust resistance in two of the three trials, while RZ323 and ISU102 in linkage group 4 had significant effect on rust resistance in one of three trials. Very significant association between loci PSB47 and TXS422 in linkage group 10 and rust resistance were detected from phenotypic data sets obtained from all three field trials.

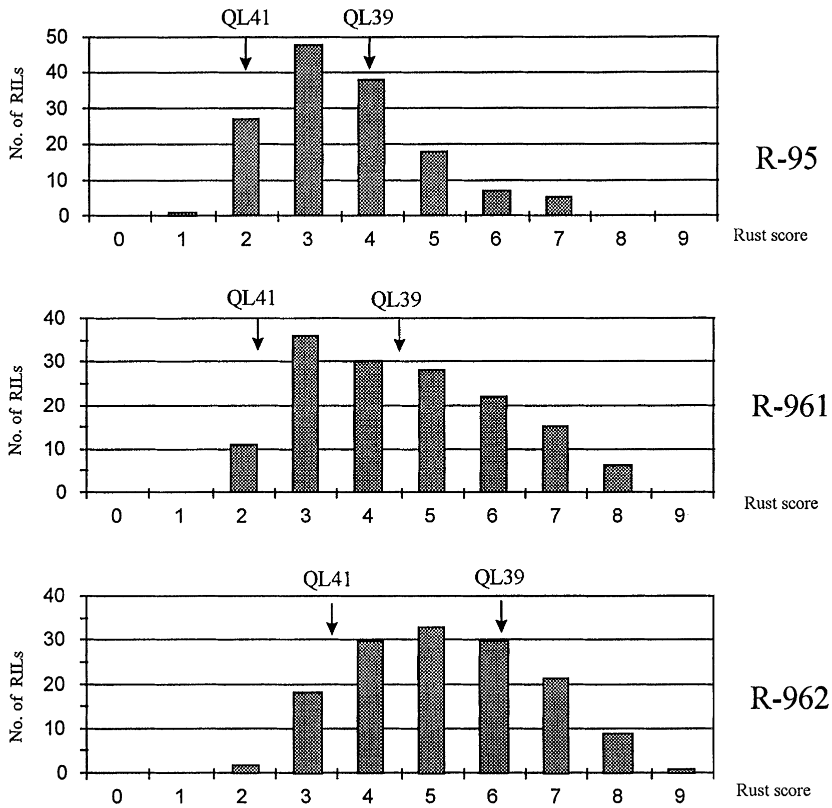

Figure 1. Histograms illustrating the frequency of rust resistance scores for the 160 RILs in three different field trials: R-95, R-961 and R-962.

For the loci in the regions of linkage group 2 and 10, the phenotypic means of genotype QL39 was greater than the phenotypic means of genotype QL41. This suggests that these regions in QL41 are associated with rust resistance. For the loci in the regions of linkage group 4 and 5, on the other hand, the phenotypic means of genotype QL39 was less than phenotypic means of genotype QL41, implying that the rust resistance has come from QL39. This is supported by the transgressive segregation noted in all data sets (Figure 1).

The percentage of the total phenotypic variation explained by each of these genomic regions varied from 6.8% to 42.6% (Table 1). The region in linkage group 10 accounted for the highest percentage of the total phenotypic variation. This result is consistent with QL41 having higher levels of rust resistance than QL39.

Discussion

Relating qualitative and quantitative inheritance

From a genetic point of view, two forms of host-pathogen interactions should be considered: 1) interactions that vary quantitatively and 2) interactions that vary qualitatively. Two types of common rust resistance have been described in sweet corn. The first one is partial resistance, which is controlled by multiple genes, shows resistance to all biotypes of pathogen *P. sorghi* and is expressed mainly at the adult plant stage. The second type of resistance is race specific resistance which is controlled by genes which act in a gene-for-gene manner with the rust fungus. These genes generally confer high level of resistance to specific rust biotypes, both at the seedling and adult stage. In sorghum, race specific resistant studies have not been reported. Investigation for rust resistance in this study was conducted in the field trials and the phenotypic scores were obtained from adult plants. The pathogens in this study, therefore, are very unlikely to be a pure race. As promising cultivars with high levels

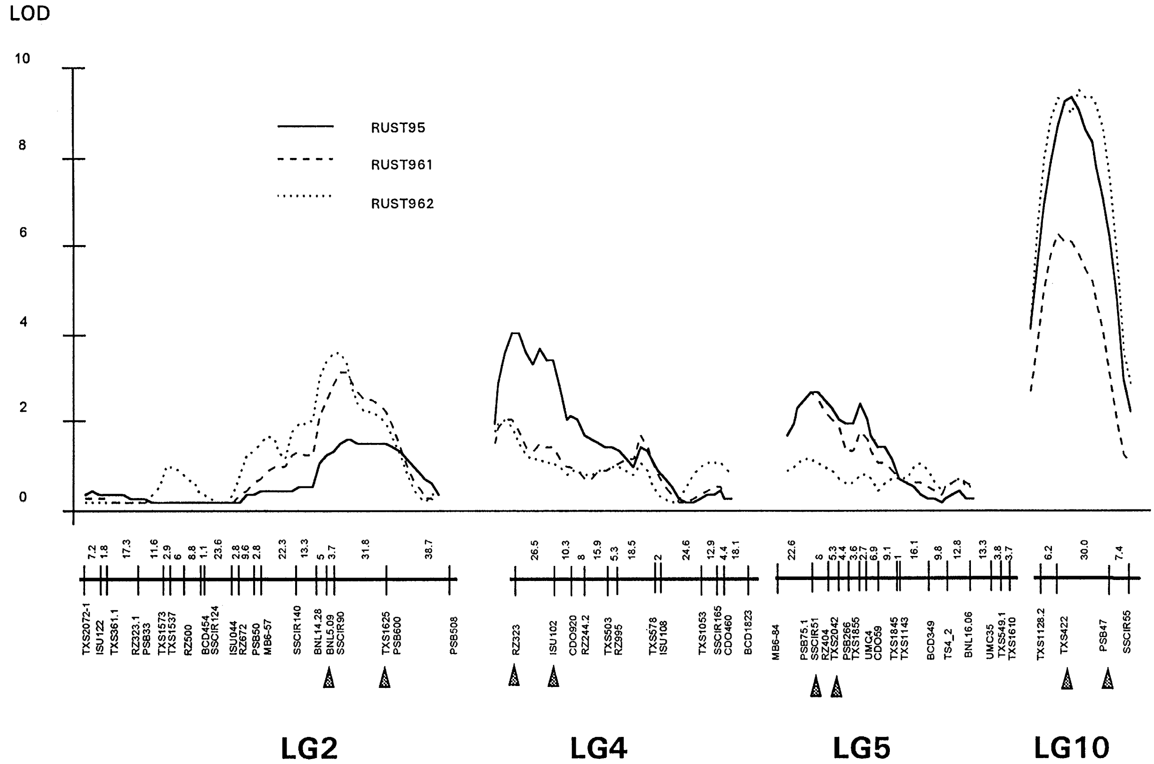

Figure 2. LOD-likelihood plots of linkage groups 2, 4 ,5 and 10 for rust resistance. Loci associated with the trait are indicated with the arrowheads.

Table 1. Locations and effects of QTL affecting rust resistance detected from both interval mapping and non-parametric mapping based on three sets of phenotypic data

Locus	LG	Interval mapping						Non-parametric mapping		
		LOD			Variation%			Level of significance		
		95	961	962	95	961	962	95	961	962
BNL5.09								**	****	******
	LG2	1.45	3.05	3.49	8.8	22.4	25.5			
TXS1625								**	*****	****
RZ323								***	**	***
	LG4	3.95	1.92	1.91	23.6	13.7	12.4			
ISU102								******	**	*
SSCIR51								****	***	**
	LG5	2.58	2.55	1.01	16.8	, 19.1	6.8			
TXS2042								****	****	*
PSB47								*******	*****	*******
	LG10	9.38	6.28	9.41	34.2	31.5	42.6			
TXS422								*******	*******	*******

Significant levels: *: 0.1; **: 0.05; ***: 0.01; ****: 0.005; *****: 0.001; ******: 0.0005; *******: 0.0001.

of race specific resistance often fail to reach their potential due to the appearance of new races, it would be much more useful in sorghum breeding to identify the genomic regions associated with resistance to different rust races.

Multiple experiments for rust evaluation, involving several field trials in two seasons, were conducted in this study. The phenotypic data were collected from field trials and rust measurements were made on a group of mature plants. The accuracy and reliability of phenotypic scoring for rust resistance is, therefore, higher than the scoring of a single F2 plant. The segregation pattern of the trait for the RIL population was observed to be a continuous distribution from all data sets, suggesting that the rust resistance is controlled by multiple genes rather than a single gene. The results of QTL mapping also indicated that there are three or four regions, which represent different genes, located on different linkage groups, which contribute to rust resistance.

Major genomic regions for rust resistance

Four genomic regions were identified consistently from at least one of the three data sets. The significant genetic effects contributing to rust resistance came from both parents of the population. The regions in linkage groups 2 and 10 came from QL41, and regions in groups 4 and 6 came from QL39. It is obvious that loci from QL41 in linkage group 10 play the major role for rust resistance in this population.

In maize, at least six loci on three chromosomes, the distal end of short arm of chromosome 10, chromosome 3 and chromosome 4, have been identified for resistance to rust (Hooker, 1985). The *Rp1* locus, which contains 14 dominant genes, is a complex rust resistance locus where multiple resistance genes are clustered (Hu and Hulbert, 1996). Genomic regions detected from this study are not associated with any regions identified for rust resistance in maize. Further investigations are needed to reveal how many genes are involved in these different regions for sorghum rust resistance, and whether the locus in group 10 is a complex rust resistance locus.

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